

BITS-PILANI, HYDERABAD

STATISTICAL DATA ANALYSIS: MAC F313

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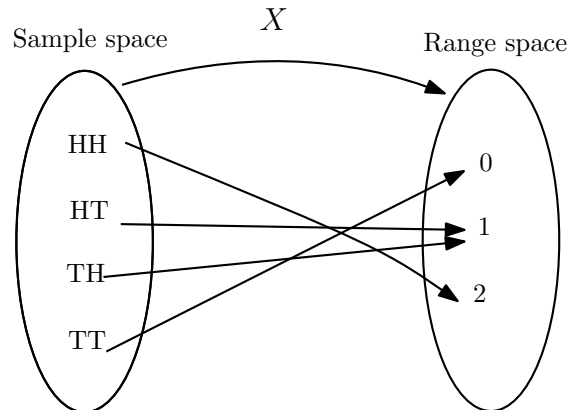
Module 2: Simulation of Random Variables-PART A



I. RANDOM VARIABLES AND PROBABILITY DISTRIBUTIONS: RECAP

A random variable is defined as a mapping X from the sample space $\mathcal{S}$ to the set of real numbers $\mathbb{R}$. (Usually random variables are denoted by capital letters X, Y, Z etc.) A random variable $X : \mathcal{S} \rightarrow \mathbb{R}$, takes any element $w \in \mathcal{S}$ to the real value $X(w) = x$. Here, the values that the random variable X takes is denoted by x . We write $X = x$ instead of $X(w) = x$. **The set of values that X takes is called a range space of X , denoted by $\mathbb{R}_X$.**

Example 2: Consider the random experiment of two fair coin tosses. Let the sample space be given by $\mathcal{S} = \{HH, HT, TH, TT\}$, where H denotes Head and T denotes Tail. Define a random variable X as the number of heads in 2 tosses. Then X can take values 0, 1, 2. Here X maps $\mathcal{S}$ to the set $\mathbb{R}_X = \{0, 1, 2\}$ which is a subset of $\mathbb{R}$. See the Figure.



There are two types of random variables based on how many values X can take (this is same as the number of elements in the range space $\mathbb{R}_X$ of X):

- Discrete random variable: X is discrete if it takes only finite or countably infinite values (that is $\mathbb{R}_X$ is finite or countably infinite) Eg: $\mathbb{R}_X = \{a_1, \dots, a_k\}$ or of the form $\mathbb{R}_X = \{a_1, a_2, \dots\}$.

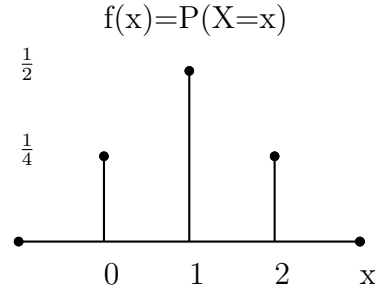


Fig. 1: pmf as a graph

$X = x$	0	1	2
$f(x) = P(X = x)$	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{1}{4}$

TABLE I: pmf as a table

- Continuous random variable: X is continuous if $\mathbb{R}_X$ is uncountable. Eg: $\mathbb{R}_X$ contains some interval $[a, b]$ or (a, b) or $\mathbb{R}_X = \mathbb{R}$.

Discrete random variable: Note that in Example 2,

$$X = 0 \leftrightarrow A = \{TT\}$$

$$X = 1 \leftrightarrow B = \{HT, TH\}$$

$$X = 2 \leftrightarrow C = \{HH\}.$$

Hence any event in the sample space is represented in terms of the random variable X . Hence one can define probability of events as

$$P(X = 0) \leftrightarrow P(A) = 1/4$$

$$P(X = 1) \leftrightarrow P(B) = 1/2$$

$$P(X = 2) \leftrightarrow P(C) = 1/4.$$

This function $P(X = x)$ for each value of x is called the **probability mass function (pmf)**, denoted by $f_X(x)$ or simply $f(x)$. For a discrete random variable, this function can be represented as in Fig. 2 and TABLE 1. **Note that for a discrete random variable, $f(x) = P(X = x)$.**

The probability mass function $f(x)$ of a discrete random variable satisfies two properties:

- $f(x) \geq 0$ for all x
- $\sum_x f(x) = 1$.

In Example 2, let us compute some probabilities,

$$P(X \leq 1) = P(X = 0) + P(X = 1) = 3/4$$

$$P(X > 1) = P(X = 2) = 1/4$$

$$P(X > 2) = 0.$$

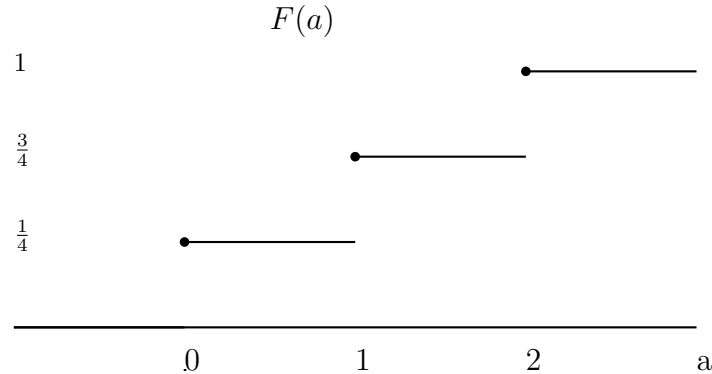
Now for a discrete random variable X , define a function called the **cumulative distribution function (CDF)**, denoted by $F_X(a)$ or simply $F(a)$ for all values of $a \in \mathbb{R}$ as

$$F(a) = P(X \leq a) = \sum_{x \leq a} f(x) = \sum_{x \leq a} P(X = x), \quad a \in \mathbb{R}.$$

In Example 7, the CDF function is given by

$$F(a) = \begin{cases} 0 & \text{for } a < 0 \\ \frac{1}{4} & \text{for } 0 \leq a < 1 \\ \frac{3}{4} & \text{for } 1 \leq a < 2 \\ 1 & \text{for } a \geq 2. \end{cases}$$

For Example 2, the graph of the CDF is given below: **Note that CDF of a discrete random variable will be a step function** like shown in the figure.



Continuous random variable: A continuous random variable is not defined at specific values. Instead, it is defined over an interval of values. For a continuous random variable X , the probability of events is calculated using **probability density function (pdf)** denoted by $f_X(x) = f(x)$ which satisfies

- $f(x) \geq 0$ for all x
- $\int_{-\infty}^{\infty} f(x)dx = 1$.

For $a < b$, the probability of the event that $a < X < b$ is calculated as

$$P(a < X < b) = \int_a^b f(x)dx.$$

Note that for a continuous random variable X ,

$$P(a < X < b) = P(a \leq X < b) = P(a < X \leq b) = P(a \leq X \leq b) = \int_a^b f(x)dx.$$

This also implies for a continuous random variable X , $P(X = a) = 0$, for any $a \in \mathbb{R}$. In addition, we have $f(x) \neq P(X = x)$.

Cumulative distribution function (CDF), $F_X(a)$ for a continuous random variable X , is defined as follows:

$$F(a) = P(X \leq a) = \int_{-\infty}^a f(x)dx, \quad a \in \mathbb{R}.$$

Survival Function or **Complementary CDF (CCDF)** is defined as

$$S(a) = 1 - F(a).$$

Example 3: : Let X be a continuous random variable with probability density function $f(x)$:

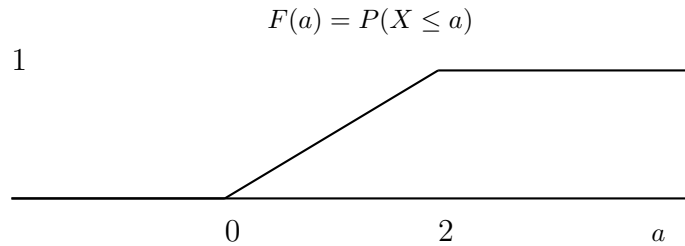
$$f(x) = \begin{cases} \frac{1}{2} & \text{for } 0 \leq x \leq 2 \\ 0 & \text{otherwise.} \end{cases}$$

Then the CDF function is given by

$$F(a) = P(X \leq a) = \begin{cases} 0 & \text{for } a < 0 \\ \frac{a}{2} & \text{for } 0 \leq a \leq 2 \\ 1 & \text{for } a > 2. \end{cases}$$

Here for $0 \leq a \leq 2$, we have $F(a) = P(X \leq a) = \int_0^a \frac{1}{2}dx = \frac{a}{2}$.

Note that CDF of a continuous random variable X will be a continuous function: (one example is shown in the figure).



Expectation and Variance of a random variable:

- Discrete random variable: Let X be a discrete random variable with pmf $f(x) = P(X = x)$. Then **expectation of X (mean)**, denoted by $E(X)$ is defined by

$$E(X) = \sum_x x f(x).$$

Here $\sum_x$ means summation runs for all possible values of the random variable X . Similarly we can define expectation of any function g of X as follows.

$$E(g(X)) = \sum_x g(x)f(x).$$

In particular,

$$E(X^2) = \sum_x x^2 f(x).$$

Letting $E(X) = \mu$, the **variance of X** , denoted by $\text{Var}(X)$ is defined as

$$\text{Var}(X) = E((X - \mu)^2) = \sum_x (x - \mu)^2 f(x).$$

- Continuous random variable: Let X be a continuous random variable with pdf $f(x)$. Then **expectation of X (mean)**, $E(X)$ is defined by

$$E(X) = \int_{-\infty}^{\infty} x f(x) dx.$$

Similarly,

$$E(X^2) = \int_{-\infty}^{\infty} x^2 f(x) dx.$$

The **variance of X** is defined as

$$\text{Var}(X) = E((X - \mu)^2) = \int_{-\infty}^{\infty} (x - \mu)^2 f(x) dx$$

- In both cases of discrete and continuous, the variance of X can be found by simpler formula given as

$$\text{Var}(X) = E(X^2) - (E(X))^2 = E(X^2) - \mu^2, \quad \text{where } \mu = E(X).$$

Properties: Important properties of expectation and variance of X (for both discrete and continuous) are given below: For constants $a, b \in \mathbb{R}$, we have

$$1) E(aX + b) = aE(X) + b.$$

Proof for continuous case:

$$\begin{aligned} E(aX + b) &= \int_{-\infty}^{\infty} (ax + b)f(x)dx \\ &= \int_{-\infty}^{\infty} axf(x)dx + \int_{-\infty}^{\infty} bf(x)dx \\ &= a \int_{-\infty}^{\infty} xf(x)dx + b \int_{-\infty}^{\infty} f(x)dx \\ &= aE(X) + b \end{aligned}$$

where the last step follows as $\int_{-\infty}^{\infty} f(x)dx = 1$.

2) $\text{Var}(aX + b) = a^2\text{Var}(X)$.

Proof for continuous case: Let $\mu = E(X)$, then we have $E(aX + b) = a\mu + b$, and

$$\begin{aligned}\text{Var}(aX + b) &= \int_{-\infty}^{\infty} [(ax + b) - (a\mu + b)]^2 f(x)dx \\ &= \int_{-\infty}^{\infty} (ax - a\mu)^2 f(x)dx \\ &= \int_{-\infty}^{\infty} a^2(x - \mu)^2 f(x)dx \\ &= a^2\text{Var}(X).\end{aligned}$$

Note that for discrete case, same proof holds by replacing integral with summation.

Moment-Generating function (MGF) of X is defined as:

- For discrete R.V. X ,

$$M_X(t) = E(e^{tX}) = \sum_x e^{tx} f(x)$$

- Continuous RV X ,

$$M_X(t) = E(e^{tX}) = \int_{-\infty}^{\infty} e^{tx} f_X(x) dx.$$

Note that

$$M_X(t) < \infty \quad \text{for all } t \in (-h, h) \text{ for some } h > 0$$

Summary

	Discrete random variable	Continuous random variable
1	R_X finite or countably infinite	R_X uncountable
2	$f(x)$: probability mass function (pmf)	$f(x)$: probability density function (pdf)
3	$f(x) = P(X = x)$ for any x	$f(x) \neq P(X = x)$
4	$P(X = a)$ may or may not be 0	$P(X = a) = 0$ for any a
5	CDF $F(x)$ is a step function	CDF $F(x)$ is a continuous function

II. LIST OF DISTRIBUTIONS

- 1) Binomial distribution
- 2) Poisson distribution
- 3) Geometric distribution
- 4) Uniform distribution

- 5) Exponential distribution
- 6) Gamma distribution
- 7) Normal distribution

Binomial distribution: Consider a random experiment with exactly two possible outcomes, “**success**” and “**failure**”. Such an experiment is called a **Bernoulli trial**. We denote the probability of success by p , then probability of failure is $1 - p$, where $0 \leq p \leq 1$.

- We repeat this Bernoulli trial n number of times.
- We assume that the trials are independent
- The probability of success is the same across all the trials.

Now define a random variable X as **the number of successes in n trials**. Here X can take values $i = 0, 1, \dots, n$. Hence this is a discrete random variable. The probability mass function $p(x)$ of X is called the **binomial distribution with parameters n and p** given by

$$p(i) = P(X = i) = \binom{n}{i} p^i (1 - p)^{n-i}, \quad \text{where} \quad \binom{n}{i} = \frac{n!}{i!(n-i)!}.$$

The binomial distribution is denoted by $B(n, p)$, and we write $X \sim B(n, p)$ which reads as “ X follows binomial distribution $B(n, p)$ ”. For a binomial random variable X , we have

$$E(X) = np$$

$$\text{Var}(X) = np(1 - p).$$

Poisson distribution: The random variable X is said to have Poisson distribution with a real parameter $\lambda > 0$ if X has the probability mass function $p(x)$ given by

$$p(i) = P(X = i) = \frac{e^{-\lambda} \lambda^i}{i!}, \quad \text{where} \quad i = 0, 1, 2, \dots$$

Here X can take countably infinite values, hence X is a discrete random variable. The Poisson distribution is denoted by $\text{Pois}(\lambda)$, and we write $X \sim \text{Pois}(\lambda)$. Here the parameter λ is called the rate parameter. Poisson random variable can be used to model given number of events occurring in a fixed interval of time (or space) if these events occur with a known constant mean rate and independently of the time since the last event. For a Poisson random variable X , we have

$$E(X) = \lambda$$

$$\text{Var}(X) = \lambda.$$

Geometric distribution: Consider a Bernoulli trial with probability of success taken as p . Then define a random variable X as **the number of Bernoulli trials to get first success**. Here X takes countably infinite

values from $1, 2, \dots$, and hence it is a discrete random variable. X follows the geometric distribution with parameter p , and we denote it as $X \sim G(p)$. Then probability mass function is given by

$$p(i) = P(X = i) = (1 - p)^{i-1}p, \quad i = 1, 2, \dots$$

The expectation and variance are given by

$$\begin{aligned} E(X) &= \frac{1}{p} \\ \text{Var}(X) &= \frac{1-p}{p^2}. \end{aligned}$$

Uniform distribution: A random variable X is said to follow the uniform distribution if the pdf of X is given by

$$f(x) = \begin{cases} \frac{1}{b-a} & a \leq x \leq b \\ 0 & \text{Otherwise.} \end{cases}$$

where $a < b$. The mean, variance, and CDF are

$$\begin{aligned} E(X) &= \frac{a+b}{2} \\ \text{Var}(X) &= \frac{1}{12}(b-a)^2 \\ F(x) &= \frac{x-a}{b-a}, \quad x \in [a, b]. \end{aligned}$$

Exponential distribution: A random variable X is said to follow an exponential distribution if the pdf of X is given by

$$f(x) = \lambda e^{-\lambda x}, \quad x \geq 0.$$

It is a continuous random variable as X takes uncountable values $[0, \infty)$. Exponential random variable can be used to model the time until an event occurs if the events occur independently and continuously over time, and the rate of occurrence λ is constant over time.

Examples: Time until a customer arrives at a service point, time until a light bulb burns out, time between phone calls at a call center, time between radioactive particle emissions.

Expectation, variance, and CDF are given by

$$\begin{aligned} E(X) &= \frac{1}{\lambda} \\ \text{Var}(X) &= \frac{1}{\lambda^2} \\ F(x) &= P(X \leq x) = 1 - e^{-\lambda x}, \quad x \geq 0. \end{aligned}$$

Gamma distribution: A random variable X is said to follow a gamma distribution if the pdf of X is given by

$$f(x) = \frac{\lambda^r}{\Gamma(r)} e^{-\lambda x} x^{r-1}, \quad x \geq 0.$$

where **Gamma Function** ($\Gamma(\alpha)$) is defined as

$$\Gamma(\alpha) = \int_0^\infty t^{\alpha-1} e^{-t} dt, \quad \alpha > 0.$$

$$\Gamma(\alpha + 1) = \alpha \Gamma(\alpha)$$

$$\Gamma(n) = (n-1)!, \quad \text{for integers } n$$

$$\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}.$$

A Gamma random variable is denote by $X \sim \Gamma(r, \lambda)$, where $r > 0$ denote the shape parameter, and $\lambda > 0$ is the rate parameter. It can be used to model the waiting time until a specified number of events occur, when the events occur independently and continuously over time, and the rate of occurrence λ is constant over time.

Examples: Time until the (r) -th customer arrives at a service point, time until the (r) -th phone call arrives at a call center, time until the (r) -th machine failure occurs, etc.

$$\begin{aligned} E(X) &= \frac{r}{\lambda} \\ \text{Var}(X) &= \frac{r}{\lambda^2}. \end{aligned}$$

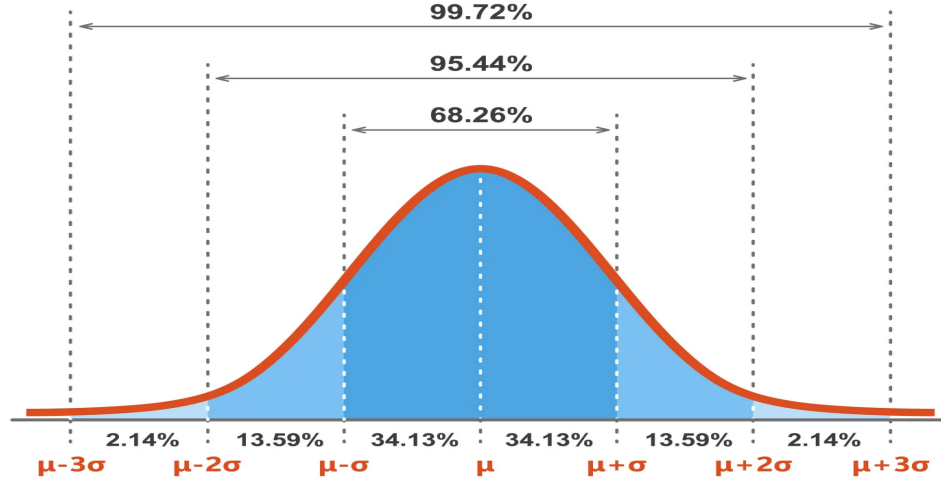
Normal distribution: The random variable X is said to have Normal distribution with real parameters $\mu \in \mathbb{R}, \sigma^2 > 0$ if X has the probability density function $f(x)$ given by

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) \quad \text{where } x \in \mathbb{R}.$$

X is a continuous random variable as $x \in \mathbb{R}$. The Normal distribution is denoted by $N(\mu, \sigma^2)$, and we write $X \sim N(\mu, \sigma^2)$. For a normal random variable X , we have

$$\begin{aligned} E(X) &= \mu \\ \text{Var}(X) &= \sigma^2. \end{aligned}$$

Here, σ is the standard deviation. For normal distribution $N(\mu, \sigma^2)$, approximately 68%, 95%, and 99.7% of the values lie within one, two, and three standard deviations (σ) of the mean (μ), respectively, as depicted in the figure.



Normal pdf is symmetric around μ , and mean, mode and median of the normal distribution are the same which is μ .

For $\mu = 0$ and $\sigma = 1$, the normal distribution $N(0, 1)$ is called the **standard normal distribution**. The standard normal random variable is usually denoted by Z , and we write $Z \sim N(0, 1)$. The pdf of the standard normal variable Z is given by

$$f(z) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{z^2}{2}\right) \quad \text{where } z \in \mathbb{R}.$$

Let random variable $X \sim N(\mu, \sigma^2)$. Then one can always convert X to the standard normal random variable Z by the change of variable $Z = \frac{X - \mu}{\sigma}$. Then it is easy to see from the properties of expectation that

$$\begin{aligned} E(Z) &= E\left(\frac{X}{\sigma} - \frac{\mu}{\sigma}\right) = \frac{1}{\sigma}E(X) - \frac{\mu}{\sigma} = 0 \\ \text{Var}(Z) &= \text{Var}\left(\frac{X}{\sigma} - \frac{\mu}{\sigma}\right) = \frac{1}{\sigma^2}\text{Var}(X) = \frac{\sigma^2}{\sigma^2} = 1. \end{aligned}$$

Hence, we have

$$X \sim N(\mu, \sigma^2) \Rightarrow Z = \frac{X - \mu}{\sigma} \sim N(0, 1).$$

For any $a > 0$, the probability $P(0 < Z < a)$ can be computed from the standard normal table, see the Z-table. Example, for $a = 1.32 = 1.3 + 0.02$, from the table first look for the row starting with 1.3, then in that row, look for the column with 0.02, the intersection value is 0.40658. Hence

$P(0 < Z < 1.32) = 0.40658$. Due to the symmetry of normal distribution, we have

$$P(Z < 0) = P(Z > 0) = 0.5 \quad (1)$$

$$P(0 < Z < a) = P(-a < Z < 0), \quad a > 0 \quad (2)$$

$$P(Z < a) = 0.5 + P(0 < Z < a), \quad a > 0. \quad (3)$$

Example 9:: Let X be a random variable which follows normal distribution with mean $\mu = 30$ and standard deviation $\sigma = 4$. Find the following probabilities: i) $P(X < 40)$ and ii) $P(X > 21)$?

Answer: Given that $X \sim N(30, 16)$. First we convert X to $Z = \frac{X - \mu}{\sigma} = \frac{X - 30}{4}$.

1) To calculate $P(X < 40)$, we have

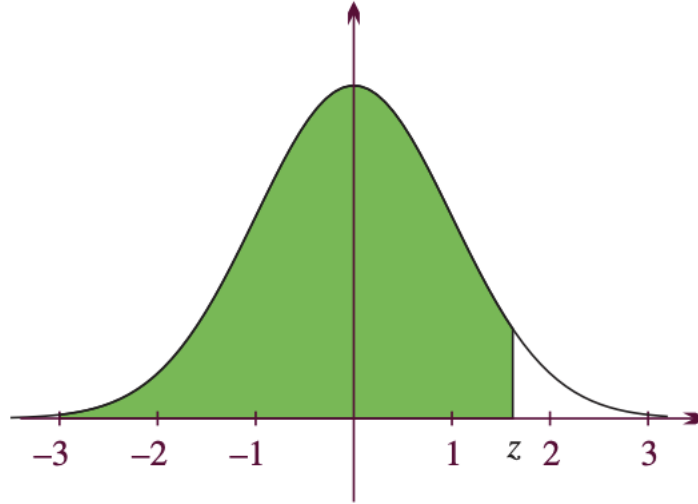
$$\begin{aligned} P(X < 40) &= P\left(\frac{X - 30}{4} < \frac{40 - 30}{4}\right) \\ &= P(Z < 2.5) \\ &= 0.5 + P(0 < Z < 2.5) \\ &= 0.5 + 0.49379 \quad (\text{value from the Z-table}) \\ &= 0.99379. \end{aligned}$$

2) To calculate $P(X > 21)$, we have

$$\begin{aligned} P(X > 21) &= P\left(\frac{X - 30}{4} > \frac{21 - 30}{4}\right) \\ &= P(Z > -2.25) \\ &= P(-2.25 < Z < 0) + 0.5 \\ &= P(0 < Z < 2.25) + 0.5 \quad (\text{from Eq (2)}) \\ &= 0.48778 + 0.5 \quad (\text{value from the Z-table}) \\ &= 0.98778. \end{aligned}$$

B. CDF Z-table for Standard Normal Distribution

Not that the CDF table can also be used to compute probabilities for the standard normal distribution $Z \sim N(0, 1)$. In the given figure, $F(z) = P(Z \leq z)$ is given by the shaded area.



This is same as $F(z) = P(Z \leq z) = 0.5 + P(0 \leq Z \leq z)$.

Example, To calculate $P(Z \leq 2.5)$, we have two methods:

1) **Using Z-table for Standard Normal Distribution for $P(0 \leq Z \leq z)$:**

$$\begin{aligned} P(Z \leq 2.5) &= 0.5 + P(0 \leq Z \leq 2.5) \\ &= 0.5 + 0.49379 \quad (\text{value from the Z-table}) \\ &= 0.99379. \end{aligned}$$

2) **Using CDF Z-table for Standard Normal Distribution:**

$$P(Z \leq 2.5) = 0.99379 \quad (\text{direct from the CDF-table}).$$

STANDARD NORMAL DISTRIBUTION: Table Values Represent AREA to the LEFT of the Z score.

Z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
-3.9	.00005	.00005	.00004	.00004	.00004	.00004	.00004	.00004	.00003	.00003
-3.8	.00007	.00007	.00007	.00006	.00006	.00006	.00006	.00005	.00005	.00005
-3.7	.00011	.00010	.00010	.00010	.00009	.00009	.00008	.00008	.00008	.00008
-3.6	.00016	.00015	.00015	.00014	.00014	.00013	.00013	.00012	.00012	.00011
-3.5	.00023	.00022	.00022	.00021	.00020	.00019	.00019	.00018	.00017	.00017
-3.4	.00034	.00032	.00031	.00030	.00029	.00028	.00027	.00026	.00025	.00024
-3.3	.00048	.00047	.00045	.00043	.00042	.00040	.00039	.00038	.00036	.00035
-3.2	.00069	.00066	.00064	.00062	.00060	.00058	.00056	.00054	.00052	.00050
-3.1	.00097	.00094	.00090	.00087	.00084	.00082	.00079	.00076	.00074	.00071
-3.0	.00135	.00131	.00126	.00122	.00118	.00114	.00111	.00107	.00104	.00100
-2.9	.00187	.00181	.00175	.00169	.00164	.00159	.00154	.00149	.00144	.00139
-2.8	.00256	.00248	.00240	.00233	.00226	.00219	.00212	.00205	.00199	.00193
-2.7	.00347	.00336	.00326	.00317	.00307	.00298	.00289	.00280	.00272	.00264
-2.6	.00466	.00453	.00440	.00427	.00415	.00402	.00391	.00379	.00368	.00357
-2.5	.00621	.00604	.00587	.00570	.00554	.00539	.00523	.00508	.00494	.00480
-2.4	.00820	.00798	.00776	.00755	.00734	.00714	.00695	.00676	.00657	.00639
-2.3	.01072	.01044	.01017	.00990	.00964	.00939	.00914	.00889	.00866	.00842
-2.2	.01390	.01355	.01321	.01287	.01255	.01222	.01191	.01160	.01130	.01101
-2.1	.01786	.01743	.01700	.01659	.01618	.01578	.01539	.01500	.01463	.01426
-2.0	.02275	.02222	.02169	.02118	.02068	.02018	.01970	.01923	.01876	.01831
-1.9	.02872	.02807	.02743	.02680	.02619	.02559	.02500	.02442	.02385	.02330
-1.8	.03593	.03515	.03438	.03362	.03288	.03216	.03144	.03074	.03005	.02938
-1.7	.04457	.04363	.04272	.04182	.04093	.04006	.03920	.03836	.03754	.03673
-1.6	.05480	.05370	.05262	.05155	.05050	.04947	.04846	.04746	.04648	.04551
-1.5	.06681	.06552	.06426	.06301	.06178	.06057	.05938	.05821	.05705	.05592
-1.4	.08076	.07927	.07780	.07636	.07493	.07353	.07215	.07078	.06944	.06811
-1.3	.09680	.09510	.09342	.09176	.09012	.08851	.08691	.08534	.08379	.08226
-1.2	.11507	.11314	.11123	.10935	.10749	.10565	.10383	.10204	.10027	.09853
-1.1	.13567	.13350	.13136	.12924	.12714	.12507	.12302	.12100	.11900	.11702
-1.0	.15866	.15625	.15386	.15151	.14917	.14686	.14457	.14231	.14007	.13786
-0.9	.18406	.18141	.17879	.17619	.17361	.17106	.16853	.16602	.16354	.16109
-0.8	.21186	.20897	.20611	.20327	.20045	.19766	.19489	.19215	.18943	.18673
-0.7	.24196	.23885	.23576	.23270	.22965	.22663	.22363	.22065	.21770	.21476
-0.6	.27425	.27093	.26763	.26435	.26109	.25785	.25463	.25143	.24825	.24510
-0.5	.30854	.30503	.30153	.29806	.29460	.29116	.28774	.28434	.28096	.27760
-0.4	.34458	.34090	.33724	.33360	.32997	.32636	.32276	.31918	.31561	.31207
-0.3	.38209	.37828	.37448	.37070	.36693	.36317	.35942	.35569	.35197	.34827
-0.2	.42074	.41683	.41294	.40905	.40517	.40129	.39743	.39358	.38974	.38591
-0.1	.46017	.45620	.45224	.44828	.44433	.44038	.43644	.43251	.42858	.42465
-0.0	.50000	.49601	.49202	.48803	.48405	.48006	.47608	.47210	.46812	.46414

List of Discrete Distributions

	Probability mass function $p(x)$	Properties
1	Binomial Distribution ($X \sim B(n, p)$) $p(i) = P(X = i) = \binom{n}{i} p^i (1 - p)^{n-i}$ where, $\binom{n}{i} = \frac{n!}{i!(n-i)!}$, and $i = 0, 1, \dots, n$.	Parameters $\rightarrow n, p$ $E(X) = np$ $\text{Var}(X) = np(1 - p).$
2	Poisson distribution ($\text{Pois}(\lambda)$) $p(i) = P(X = i) = \frac{e^{-\lambda} \lambda^i}{i!} \quad i = 0, 1, \dots$	Parameter $\rightarrow \lambda$ $E(X) = \lambda$ $\text{Var}(X) = \lambda$
3	Geometric distribution ($G(p)$) $p(i) = P(X = i) = (1 - p)^{i-1} p, \quad i = 1, 2, \dots$	Parameter $\rightarrow p$ $E(X) = \frac{1}{p}$ $\text{Var}(X) = \frac{1 - p}{p^2}$

List of Continuous Distributions

	pdf $f(x)$ with domain	Properties
1	Uniform Distribution ($X \sim \text{Uniform}(a, b)$) $f(x) = \begin{cases} \frac{1}{b-a} & a \leq x \leq b \\ 0 & \text{Otherwise.} \end{cases}$ Here, $a < b$.	$E(X) = \frac{a+b}{2}$ $\text{Var}(X) = \frac{1}{12}(b-a)^2$ $F(x) = \frac{x-a}{b-a}, \quad x \in [a, b].$
2	Exponential Distribution ($\text{Exp}(\lambda)$) $f(x) = \lambda e^{-\lambda x}, \quad x \geq 0$ Parameter: $\lambda > 0$ (rate)	Notation $X \sim \text{Exp}(\lambda)$ $E(X) = \frac{1}{\lambda}$ $\text{Var}(X) = \frac{1}{\lambda^2}$ $F(x) = 1 - e^{-\lambda x}, \quad x \geq 0.$
	Gamma Distribution ($\Gamma(r, \lambda)$) $f(x) = \frac{\lambda^r}{\Gamma(r)} e^{-\lambda x} x^{r-1}, \quad x > 0.$ Parameters: $\lambda > 0$ (rate), $r > 0$ (shape)	Notation $X \sim \Gamma(\lambda, r)$ $E(X) = \frac{r}{\lambda}$ $\text{Var}(X) = \frac{r}{\lambda^2}.$
4	Normal Distribution ($N(\mu, \sigma^2)$) $f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \quad x \in \mathbb{R}.$ $\mu \in \mathbb{R}$ (location), $\sigma^2 > 0$ (variance)	Notation $X \sim N(\mu, \sigma^2)$ $E(X) = \mu$ $\text{Var}(X) = \sigma^2$ $F(x) = \Phi\left(\frac{x-\mu}{\sigma}\right).$